

Economics 114 A2 Feedback

Question 39:

Most students got the Nash equilibrium

Most students also identified what the socially optimal outcome was, however, the motivation was not always clear. It was important to indicate that there was a pareto improvement when moving from the Nash Equilibrium to the Socially optimal outcome. In the socially optimal outcome, both players were better off than in the Nash Equilibrium.

Question 40.1:

A fair number of students answered this question successfully. The question could be answered from multiple angles, here are few: Points B and C fall on a pareto efficiency curve, so neither point can dominate the other. Beginning at allocation B, moving to allocation C does not bring about a pareto improvement since Mpho is better off at point C than at point B; but Sipho is worse off.

Question 40.2:

An increase in Mpho's bargaining power would be required.

Question 41

For section 41.2, make sure you understand how to calculate efficiency units per rand (e/w).

Question 42:

To determine whether the price ceiling is binding, you need to calculate the equilibrium price. The equilibrium price is where the demand curve intersects the supply curve. Use the inverse supply and demand functions to calculate the equilibrium quantity. Then, use that quantity to determine the price (P). If the market price is above the ceiling price, the price ceiling is non-binding. There is no need to determine the surplus or shortage, as this is not asked.

Question 43:

If you use the equation $Elasticity = -P/Q \times 1/slope$, you need to determine Q by using the demand function $Q_d = 50 - 0.4P$, where $P = R40$. Now you have both P and Q. For the slope, rewrite the equation to the inverse demand function. The slope is therefore 2.5 and not 0.4.

Since the elasticity is lower than 1, the demand is inelastic. Therefore, you should advise Susan not to lower her price, as this will decrease total revenue (TR). The increase in quantity demanded will not offset the decrease in price

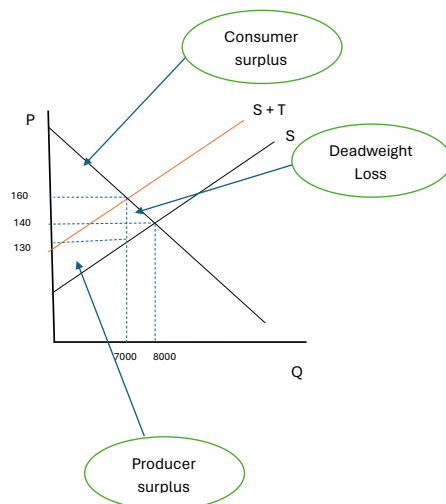
Question 44.1:

Most students were able to arrive at the correct answer. Recall the condition for market equilibrium, $Q_d = Q_s$, and then solve for P and Q.

Question 44.2:

Here students need to add the tax on supply and then re-solve for P and Q using $Q_d = Q_s + \text{Tax}$ as the equilibrium condition.

Question 44.3:



Question 44.4:

Some students made silly mistakes with this question. The tax per unit was already given in the question as R30 per unit. In question 44.2 we worked out the quantity bought as 7000, so tax revenue is R210.

Question 45

Many students did not identify that the Pareto efficient quantity is calculated by $MSB = MSC$. Some students set up the equation correctly, but then their algebra let them down and they calculated the wrong answer.

About half the students were able to calculate the correct tax rate of 300 [$\text{Tax} = \text{MEC}$]. Only a few students then worked out the tax revenue by multiplying the Tax by the Quantity

The last question was answered very poorly. Students were vague in their answers and made general statements about “they must pay them” without specifying that they is (the “loosing party”) and them is (the producers). Students also did not specify why the loosing party should pay the producers (to lower production) and how much they must pay (up to MEC per ton not produced and the producer willing to accept $P-MPC$ or more per ton to reduce production... or to cover lost profits by decreasing production to the socially optimal amount)